

Quick Reference Cheat Sheet

This page summarizes the most important commands and tips from the guide.

RStudio Shortcuts

Action	Windows	Mac
Run current line	Ctrl + Enter	Cmd + Enter
Save file	Ctrl + S	Cmd + S
Find text	Ctrl + F	Cmd + F
Comment/uncomment code	Ctrl + Shift + C	Cmd + Shift + C

Important RStudio Panes

Pane	Purpose
Source	Write and save code
Console	Run commands and see output
Environment	See objects you created
Files	Find files and folders
Plots	View graphs
Packages	Manage add-on tools
Help	Search for help

Common R Commands

Goal	Command
Add numbers	2 + 3
Store a value	x <- 10
Print a value	print(x)
Check working directory	getwd()
Check if a file exists	file.exists("mydata.csv")
Import CSV	data <- read.csv("mydata.csv")
View first rows	head(data)
View structure	str(data)
Summarize data	summary(data)
View column names	names(data)
Calculate a mean	mean(data\$quiz_score)

Goal	Command
Make a frequency table	<code>table(data\$major)</code>
Make a plot	<code>plot(data\$study_hours, data\$quiz_score)</code>

File Management Tips

- Keep your script and data file in the same project folder.
- Use clear file names.
- Avoid spaces in file names.
- Use underscores `_` instead of spaces.
- Avoid saving important files only in Downloads.
- Check your working directory with `getwd()`.
- Check whether R can find your file with `file.exists()`.

Good File Names

Good:

```
first_script.R
mydata.csv
lab_report.qmd
```

Avoid:

```
Untitled1.R
my data final FINAL.csv
lab report new version 2.qmd
```

Importing Data

Use:

```
data <- read.csv("mydata.csv")
```

Then inspect it:

```
head(data)
str(data)
summary(data)
names(data)
```

Simple Analysis Examples

Mean:

```
mean(data$quiz_score)
```

Frequency table:

```
table(data$major)
```

Scatterplot:

```
plot(
  data$study_hours,
  data$quiz_score,
  main = "Study Hours and Quiz Score",
  xlab = "Study Hours",
  ylab = "Quiz Score"
)
```

Histogram:

```
hist(
  data$quiz_score,
  main = "Distribution of Quiz Scores",
  xlab = "Quiz Score"
)
```

Common Errors and Quick Fixes

Problem	Possible Fix
File not found	Check <code>getwd()</code> and <code>file.exists("mydata.csv")</code>
Object not found	Run the code that creates the object
Typing mistake	Check spelling and capitalization
Missing quote	Put file names in quotation marks
Missing parenthesis	Make sure every (has a matching)
Rendering error	Run the document from top to bottom and check earlier code

Problem	Possible Fix
Package problem	Check spelling, install the package, then load it with <code>library()</code>

File Not Found Checklist

If R cannot find your CSV file, try this:

```
getwd()
file.exists("mydata.csv")
```

If `file.exists("mydata.csv")` gives:

```
[1] FALSE
```

then R cannot find the file in the current folder.

Check that:

- The file is saved in your project folder.
- The file name is exactly `mydata.csv`.
- The file is not accidentally named `mydata.csv.txt`.
- You are working in the correct folder.

Object Names

Good object names:

```
quiz_score <- 90
study_hours <- 6
my_data <- read.csv("mydata.csv")
```

Avoid object names with spaces:

```
quiz score <- 90
study hours <- 6
my data <- read.csv("mydata.csv")
```

Useful Beginner Habits

- Save your script often.
- Run code one small piece at a time.
- Read error messages slowly.
- Check spelling carefully.
- Keep your data and script in the same folder.
- Write short comments to explain your code.
- Explain your results in sentences.

A Complete First Workflow

Here is a simple workflow from start to finish:

```
# Check where R is looking for files
getwd()

# Check whether the data file exists
file.exists("mydata.csv")

# Import the data
data <- read.csv("mydata.csv")

# Look at the data
head(data)
str(data)
summary(data)

# Calculate results
mean(data$quiz_score)
table(data$major)

# Make a plot
plot(
  data$study_hours,
  data$quiz_score,
  main = "Study Hours and Quiz Score",
  xlab = "Study Hours",
  ylab = "Quiz Score",
  pch = 19,
  col = "steelblue"
)
```

Final Reminder

You now know how to complete a first basic R workflow:

1. Open RStudio.
2. Create a script.
3. Write code.
4. Save your work.
5. Organize your files.
6. Import a CSV dataset.
7. Inspect the data.
8. Calculate simple results.
9. Explain what the results mean.

That is a successful first R workflow!